

PPA Results (65nm)

Table6: Processor Area.

Metric	Generic	Map	Incremental	Final
Cell Area	13,94,233	1,164,744	1,160,756	1,160,756
Total cell area	13,94,233	1,164,744	1,160,756	1,160,756
Leaf instances	114,980	58,394	56,052	56,052
Total instances	114,980	58,394	56,052	56,052

Chip Area:
1,160,756 μm^2

Table7: Processor Timing.

Metric	Generic	Map	Incremental	Final
Slack (ps)	11,447	580	54	54
• R2R	14,135	10,369	9,750	9,750
• I2R	17,843	14,399	13,974	13,974
• R2O	11,447	580	54	54
• I2O	15,155	4,611	4,278	4,278

Clock Frequency:
52Mhz

Power: 89.456 mW

Comparisons of Application RISC-V Cores (22FDX ASIC)

Table8: Comparison with other available designs.

Design	Area (mm ²)	Power(mW) @ (0.85V, 50MHz)	Performance
Rocket Core	0.22	15.76	813MHz
BOOM	0.92	139.03	943MHz (0.9V) 70MHz (0.6V)
CVA6	0.52	26.30	738MHz
SHAKTI	0.84	23.81	685MHz
RISCV(This design), 65nm	1.16	89.456	52Mhz

1. The core area and power will reduce as we go to lower tech nodes like 45nm or 28nm.
2. Performance will increase since the cell delays will be less.

Comparisons with Target Specification

Table9: Target Specification.

	Area (mm ²)	Power (mW)	Performance(MHz)
RV32G (Our design)	5	30	200

Table10: Achieved Specification.

Design	Area (mm ²)	Power(mW) @ (0.85V, 50MHz)	Performance
RISCV(This design), 65nm	1.16	89.456	52MHz

1. Performance bottleneck due to long memory path and RO (Register-to-output) paths.
2. Power is a factor of switching activity and voltage/performance